

연습문제 1.5 - 비선형상미분방정식 (Bernoulli · 치환)

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$$y' + xy = y^{-1}, \quad y(0) = 3$$

베르누이 · 초기값

Step 1 표준형으로 정리 (베르누이 방정식인지 판별)

$$y' + p(x)y = g(x)y^a \quad p(x) = x, \quad g(x) = 1, \quad a = -1 : a \neq 0, 1 \text{ 이므로 비선형 (Bernoulli)}$$

Step 2 치환하여 선형화

$$u = y^{1-a} = y^{1-(-1)} = y^2, \quad u' = 2yy' = 2y(-xy + y^{-1}) = -2xy^2 + 2 = -2xu + 2 \\ \Rightarrow u' + 2xu = 2 \quad (u \text{에 관한 선형 비제차}) \quad p(x) = 2x, \quad r(x) = 2$$

Step 3 일반해 구하기

$$h = \int p(x) dx = \int 2x dx = x^2 \\ u = e^{-h} \left[\int e^h r dx + c \right] = e^{-x^2} \left[2 \int e^{x^2} dx + c \right] \\ = e^{-x^2} \left[2 \int_0^x e^{t^2} dt + c \right] \quad (\int e^{x^2} dx \text{는 초등함수가 아님} \rightarrow \text{정적분, 상수를 } c \text{로 다시 둠})$$

$$\therefore \text{일반해: } y^2 = e^{-x^2} \left(2 \int_0^x e^{t^2} dt + c \right)$$

Step 4 특수해 구하기

초기조건 적용

$$u(0) = y(0)^2 = 9 = e^0 (2 \cdot 0 + c) \Rightarrow c = 9$$

$$\text{특수해: } y^2 = e^{-x^2} \left(9 + 2 \int_0^x e^{t^2} dt \right)$$

$$\text{즉 } y = e^{-x^2/2} \sqrt{9 + 2 \int_0^x e^{t^2} dt}$$

$y(0) = 3 > 0$ 이므로 양의 근, $9 + 2 \int_0^x e^{t^2} dt > 0$ 인 0을 포함하는 구간

Step 5 검증

$$u = e^{-x^2} \left(9 + 2 \int_0^x e^{t^2} dt \right) \Rightarrow u' = -2xe^{-x^2} \left(9 + 2 \int_0^x e^{t^2} dt \right) + e^{-x^2} \cdot 2e^{x^2} = -2xu + 2 \\ y' = \frac{u'}{2y} = \frac{-2xu + 2}{2y} = -\frac{xy^2}{y} + \frac{1}{y} = -xy + y^{-1} \Rightarrow y' + xy = y^{-1} \quad \checkmark \quad y(0) = \sqrt{9} = 3 \quad \checkmark$$

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$$2xyy' + (x-1)y^2 = x^2e^x \quad (y^2 = z \text{로 놓아라})$$

치환 $z = y^2$ · 선형

Step 1 치환하여 표준형으로 정리

$$z = y^2, \quad z' = 2yy' \Rightarrow xz' + (x-1)z = x^2e^x \\ z' + \left(1 - \frac{1}{x}\right)z = xe^x \quad p(x) = 1 - \frac{1}{x}, \quad r(x) = xe^x : z \text{에 관한 선형 비제차}$$

Step 2 일반해 구하기

$$h = \int p(x) dx = \int \left(1 - \frac{1}{x}\right) dx = x - \ln|x| \quad (x > 0 \text{ 구간에서}) \quad e^h = \frac{e^x}{x}, \quad e^{-h} = xe^{-x} \\ z = e^{-h} \left[\int e^h r dx + c \right] = xe^{-x} \left[\int \frac{e^x}{x} \cdot xe^x dx + c \right] = xe^{-x} \left[\int e^{2x} dx + c \right] = xe^{-x} \left[\frac{1}{2}e^{2x} + c \right] \\ \Rightarrow z = \frac{1}{2}xe^{2x} + cxe^{-x}$$

$$\therefore \text{일반해: } y^2 = \frac{1}{2}xe^{2x} + cxe^{-x} = x \left(\frac{1}{2}e^{2x} + ce^{-x} \right) \quad (x \neq 0, \text{ 우변 } > 0 \text{인 구간})$$

Step 3 검증

$$z' = \frac{1}{2}e^{2x} + \frac{1}{2}xe^{2x} + ce^{-x} - cxe^{-x} \\ xz' = \frac{1}{2}xe^{2x} + \frac{1}{2}x^2e^{2x} + cxe^{-x} - cx^2e^{-x}, \quad (x-1)z = \frac{1}{2}x^2e^{2x} - \frac{1}{2}xe^{2x} + cx^2e^{-x} - cxe^{-x} \\ xz' + (x-1)z = x^2e^x \quad \checkmark \\ 2xyy' = xz' \text{ 이므로 } 2xyy' + (x-1)y^2 = x^2e^x \quad \checkmark$$

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$$xy'' + 2y' + xy = 0, \quad y_1 = \frac{\cos x}{x}$$

차수축소법

Step 1 표준형으로 정리

$$y'' + \frac{2}{x}y' + y = 0 \quad p(x) = \frac{2}{x}, \quad q(x) = 1$$

Step 2 차수축소법 적용

$$\int p dx = \int \frac{2}{x} dx = 2 \ln |x|, \quad e^{-\int p dx} = e^{-2 \ln |x|} = \frac{1}{x^2}, \quad y_1^2 = \frac{\cos^2 x}{x^2}$$

$$U = \frac{1}{y_1^2} e^{-\int p dx} = \frac{x^2}{\cos^2 x} \cdot \frac{1}{x^2} = \sec^2 x, \quad u = \int U dx = \int \sec^2 x dx = \tan x$$

Step 3 두 번째 해 구하기

$$y_2 = u y_1 = \tan x \cdot \frac{\cos x}{x} = \frac{\sin x}{x}$$

$$\therefore \text{기저: } y_1 = \frac{\cos x}{x}, \quad y_2 = \frac{\sin x}{x}$$

$$\text{일반해: } y = \frac{c_1 \cos x + c_2 \sin x}{x} \quad (x \neq 0)$$

Step 4 검증

$$y_2 = \frac{\sin x}{x}, \quad y_2' = \frac{x \cos x - \sin x}{x^2}, \quad y_2'' = \frac{-x^2 \sin x - 2x \cos x + 2 \sin x}{x^3}$$

$$x y_2'' + 2 y_2' + x y_2 = \frac{-x^2 \sin x - 2x \cos x + 2 \sin x}{x^2} + \frac{2x \cos x - 2 \sin x}{x^2} + \sin x$$

$$= -\sin x + \sin x = 0 \quad \checkmark$$

$$\frac{y_2}{y_1} = \tan x \neq \text{const} \implies y_1, y_2 \text{ 1차독립 (기저)} \quad \checkmark$$

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$$x^2 y'' + x y' - 4y = 0, \quad y_1 = x^2$$

차수축소법

Step 1 표준형으로 정리

$$y'' + \frac{1}{x}y' - \frac{4}{x^2}y = 0 \quad p(x) = \frac{1}{x}, \quad q(x) = -\frac{4}{x^2}$$

Step 2 차수축소법 적용

$$\int p dx = \int \frac{1}{x} dx = \ln |x| \quad (x > 0 \text{ 구간에서}) \quad e^{-\int p dx} = e^{-\ln x} = \frac{1}{x}, \quad y_1^2 = x^4$$

$$U = \frac{1}{y_1^2} e^{-\int p dx} = \frac{1}{x^4} \cdot \frac{1}{x} = x^{-5}, \quad u = \int U dx = \int x^{-5} dx = -\frac{1}{4} x^{-4}$$

Step 3 두 번째 해 구하기

$$y_2 = u y_1 = -\frac{1}{4} x^{-4} \cdot x^2 = -\frac{1}{4} x^{-2} \implies \text{상수배를 떼어 } y_2 = x^{-2} = \frac{1}{x^2}$$

$$\therefore \text{기저: } y_1 = x^2, \quad y_2 = \frac{1}{x^2}$$

$$\text{일반해: } y = c_1 x^2 + \frac{c_2}{x^2} \quad (x \neq 0)$$

Step 4 검증

$$y_2 = x^{-2}, \quad y_2' = -2x^{-3}, \quad y_2'' = 6x^{-4}$$

$$x^2 y_2'' + x y_2' - 4y_2 = 6x^{-2} - 2x^{-2} - 4x^{-2} = 0 \quad \checkmark$$

$$\frac{y_2}{y_1} = x^{-4} \neq \text{const} \implies y_1, y_2 \text{ 1차독립 (기저)} \quad \checkmark$$

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$$y'' + 4y' + (\pi^2 + 4)y = 0$$

경우 3 · 공액복소근

Step 1 특성방정식

$$y = e^{\lambda x} \implies \lambda^2 + 4\lambda + (\pi^2 + 4) = 0$$

Step 2 근의 판별

$$a = 4, b = \pi^2 + 4 : \quad a^2 - 4b = 16 - 4\pi^2 - 16 = -4\pi^2 < 0 \implies \text{경우 3 (공액복소근)}$$

$$\lambda = -\frac{a}{2} \pm i\omega, \quad \omega = \sqrt{b - \frac{a^2}{4}} = \sqrt{\pi^2 + 4 - 4} = \pi \implies \lambda = -2 \pm i\pi$$

Step 3 일반해 구하기

$$\therefore \text{일반해: } y = e^{-2x} (A \cos \pi x + B \sin \pi x)$$

Step 4 검증

$$u = A \cos \pi x + B \sin \pi x \text{ 로 놓으면 } u'' = -\pi^2 u, \quad y = e^{-2x} u$$

$$y' = e^{-2x} (u' - 2u), \quad y'' = e^{-2x} (u'' - 4u' + 4u)$$

$$y'' + 4y' + (\pi^2 + 4)y = e^{-2x} [u'' - 4u' + 4u + 4u' - 8u + (\pi^2 + 4)u] = e^{-2x} [u'' + \pi^2 u] = e^{-2x} [-\pi^2 u + \pi^2 u] = 0$$

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$$y'' + y' + 3.25y = 0$$

경우 3 · 공액복소근

Step 1 특성방정식

$$\lambda^2 + \lambda + 3.25 = 0$$

Step 2 근의 판별

$$a = 1, b = 3.25 : \quad a^2 - 4b = 1 - 13 = -12 < 0 \implies \text{경우 3 (공액복소근)}$$

$$\omega = \sqrt{b - \frac{a^2}{4}} = \sqrt{3.25 - 0.25} = \sqrt{3} \implies \lambda = -\frac{1}{2} \pm i\sqrt{3}$$

Step 3 일반해 구하기

$$\therefore \text{일반해: } y = e^{-x/2} (A \cos \sqrt{3} x + B \sin \sqrt{3} x)$$

Step 4 검증

$$u = A \cos \sqrt{3} x + B \sin \sqrt{3} x, \quad u'' = -3u, \quad y = e^{-x/2} u$$

$$y' = e^{-x/2} (u' - \frac{1}{2}u), \quad y'' = e^{-x/2} (u'' - u' + \frac{1}{4}u)$$

$$y'' + y' + 3.25y = e^{-x/2} [u'' - u' + \frac{1}{4}u + u' - \frac{1}{2}u + 3.25u] = e^{-x/2} [u'' + 3u] = e^{-x/2} [-3u + 3u] = 0 \quad \checkmark$$

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$$y'' + 2k^2 y' + k^4 y = 0$$

경우 2 · 실이중근

Step 1 특성방정식

$$\lambda^2 + 2k^2 \lambda + k^4 = 0$$

Step 2 근의 판별

$$a = 2k^2, b = k^4 : \quad a^2 - 4b = 4k^4 - 4k^4 = 0 \implies \text{경우 2 (실이중근)}$$

$$\lambda^2 + 2k^2 \lambda + k^4 = (\lambda + k^2)^2 = 0 \implies \lambda = -k^2 \text{ (이중근)}$$

Step 3 일반해 구하기

$$\therefore \text{일반해: } y = (c_1 + c_2 x) e^{-k^2 x}$$

Step 4 검증

$$y = (c_1 + c_2 x) e^{-k^2 x}, \quad y' = e^{-k^2 x} [c_2 - k^2 (c_1 + c_2 x)]$$

$$y'' = e^{-k^2 x} [-k^2 c_2 - k^2 c_2 + k^4 (c_1 + c_2 x)] = e^{-k^2 x} [-2k^2 c_2 + k^4 (c_1 + c_2 x)]$$

$$y'' + 2k^2 y' + k^4 y = e^{-k^2 x} [-2k^2 c_2 + k^4 (c_1 + c_2 x) + 2k^2 c_2 - 2k^4 (c_1 + c_2 x) + k^4 (c_1 + c_2 x)] = 0 \quad \checkmark$$

Step 1 특성방정식

$$\lambda^2 + 3 = 0$$

Step 2 근의 판별

$$a = 0, b = 3: \quad a^2 - 4b = -12 < 0 \implies \text{경우 3 (공액복소근)}, \quad \lambda = \pm i\sqrt{3}$$

Step 3 일반해 구하기 (연산자 인수분해)

$$D^2 + 3I = (D - i\sqrt{3}I)(D + i\sqrt{3}I) \quad (\text{실수 계수의 일차 연산자로는 인수분해되지 않으므로 복소 인수로})$$

$$(D - i\sqrt{3}I)y_1 = 0 \implies y_1 = e^{i\sqrt{3}x}, \quad (D + i\sqrt{3}I)y_2 = 0 \implies y_2 = e^{-i\sqrt{3}x}$$

$$\text{두 인수는 교환되므로 } (D^2 + 3I)y_1 = (D + i\sqrt{3}I)(D - i\sqrt{3}I)y_1 = 0, \quad (D^2 + 3I)y_2 = 0$$

$$\cos \sqrt{3}x = \frac{1}{2}(y_1 + y_2), \quad \sin \sqrt{3}x = \frac{1}{2i}(y_1 - y_2) \implies \text{실수 기저 } \cos \sqrt{3}x, \sin \sqrt{3}x$$

$$\therefore \text{일반해: } y = A \cos \sqrt{3}x + B \sin \sqrt{3}x$$

Step 4 검증

$$y' = -\sqrt{3}A \sin \sqrt{3}x + \sqrt{3}B \cos \sqrt{3}x, \quad y'' = -3A \cos \sqrt{3}x - 3B \sin \sqrt{3}x = -3y$$

$$(D^2 + 3I)y = y'' + 3y = -3y + 3y = 0 \quad \checkmark$$